GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS
PHASE 6: Performance Testing

Project theme: historical Olympic participation and medal performance.

Objective

Verify data quality, calculation accuracy, usability, and reasonable processing performance.

Data validation checks

- Confirm the expected columns are present.
- Record total row count and inspect the year range.
- Check null values in Age, Height, Weight, and Medal where applicable.
- Confirm Year is numeric and Medal categories are recognized.
- Check for unexpected duplicate rows without assuming repeated athletes are errors.

Calculation tests

- Verify medal summaries include only Gold, Silver, and Bronze.
- Compare grouped totals with filtered source rows.
- Check that total counts reconcile when groups are exhaustive.
- Manually validate a small sample for one year, NOC, or sport.
- Confirm blank medals are not counted as awards.

Visualization tests

- Chart titles identify the measure and time period.
- Axes and legends are readable and correctly labelled.
- Filters, where used, update summaries consistently.
- Charts do not imply causation from descriptive trends.
- Missing-data and comparability notes are visible.

Acceptance criteria

- All core columns load without parsing errors.
- Summary tables are reproducible.
- Counts are checked against source records.
- Reduced CSV remains below 25 MB.
- Known limitations are documented before presentation.

Expected phase deliverable

Evidence of data and calculation validation.